1 Check list

Here we will collect the current task:

- Getting familiar with general framework for rotational-free shells.
  - identification of the problem,
  - variational formulation,
  - large deformation framework - continuum mechanics,
  - getting familiar with with nonlinear problems - linearization of kinematic quantites,
- Numerical implementation of the basic integration in discretized setting.
- Implementation of the solver for the Poisson problem.
- Nonlinear problems.
- Kirchhoff-Love membranes.

2 Integration

The purpose of this task it to implement the general integration routine for a function using Lagrange elements. It is important to get familiar how the discretized domain reproduces the data. To achieve this considering the domain $\mathcal{S}$ it can be subdivided into elements as

$$\mathcal{S} = \bigcup_{\mathcal{S}^e \in \mathcal{S}} \mathcal{S}^e \ , \tag{1}$$

where $\mathcal{S}^e$ defines finite element, with $e = [1, 2, \ldots, n_{\mathrm{el}}]$ and n_{el} being the number of elements. The discreate domain can be represented as

$$\boldsymbol{x}\left(\boldsymbol{\xi}\right) = \sum_{A=1}^{n_e} N_A\left(\boldsymbol{\xi}\right) \boldsymbol{x}_A = \mathbf{N}_e \mathbf{x}_e \ , \tag{2}$$

where $\mathbf{N}_e = [N_1\mathbf{I},\, N_2\mathbf{I},\, \ldots,\, N_{n_e}\mathbf{I}]$ and $\mathbf{x}_e = [\boldsymbol{x}_1,\, \boldsymbol{x}_2,\, \ldots,\, \boldsymbol{x}_{n_e}]$ denotes element shape function and element node arrays with n_e being number of element nodes. Note that the N_A are linearly independent basis functions. In similar fashion the arbitrary function $f\left(\boldsymbol{\xi}\right)$ can be approximated by a polynomial $f_p\left(\boldsymbol{\xi}\right)$ as

$$f\left(\boldsymbol{\xi}\right) \approx f\left(\boldsymbol{\xi}\right) = \sum_{A=1}^{n_e} N_A\left(\boldsymbol{\xi}\right) c_A \ , \tag{3}$$

where c_A are unknown coefficients which can be determined by different approximation methods such as interpolation of L_2 projection. To use the former, it is possible to write following system of equations

$$\mathbf{A}\mathbf{c} = \mathbf{f} \tag{4}$$

with

$$A_{ij} = N_i\left(\xi_j\right) \quad \text{and} \quad f_i = f\left(\xi_i\right) \ . \tag{5}$$

The most comonly used approximation function are the Lagrange polynomials. These can be defined as

$$N_A\left(\xi\right) = \prod_{\substack{k=1 \\ k \neq i}}^{n} \frac{\xi - \xi_k}{\xi_i - \xi_k} \ , \tag{6}$$

where j^{th} basis of order $p = n - 1$ is spawned by n point ξ_k. Linear combinations of of Lagrange basis functions are used to create approximation spaces in finite element setting. To test the approximation using Lagrange polynomials consider following function

$$f\left(\xi\right) = \sin\left(5\pi\xi\right) + \frac{1}{\left(1.05 - \xi\right)} \ . \tag{7}$$

Using equations (4) and (5) it is possible to calculate constant coefficients c_A and then visualize approximation using (3).

3 FEM - Introduction

Heat equation is a potential problem described by Laplace equation for homogeneous problem or by Poisson equation. Defining the boundary value problem (BVP)

$$\begin{cases} \nabla^2 u = s & \text{in } \mathcal{D} \ , \\ u = \bar{u} & \text{on } \mathcal{S}_u \ , \\ q = \frac{\partial u}{\partial \boldsymbol{x}} \cdot \boldsymbol{n} = \bar{q} & \text{on } \mathcal{S}_q \ , \end{cases} \tag{8}$$

where u defines the unknown potential, $\bar{u}$ is the prescribed value of potential on $\mathcal{S}_u$ portion of the boundary and $\bar{q}_q$ is prescribed flux on $\mathcal{S}_q$ with $\mathcal{S} = \mathcal{S}_u + \mathcal{S}_q$. The source term is denoted as s. Eq. 8 requires that u is twice differentiable. This is called that given problem is defined in a strong form. To reduce this requirement it is possible to derive the weak form of the problem. To achieve this it is first necessary to define two functional spaces, $\mathcal{V}$ composed of test functions as

$$\mathcal{V} = \{w \in H^1(\mathcal{D}) \,|\, w = 0 \text{ on } \mathcal{S}_u\} \tag{9}$$

and $\mathcal{W}$ defining the collection of function called trial solutions which can be written as

$$\mathcal{W} = \{u \in H^1(\mathcal{D}) \,|\, u = \bar{u} \text{ on } \mathcal{S}_u\} \ . \tag{10}$$

The first step in weak form derivation by means of weighted residual method is to multiply our model problem by weighting function $w = \mathcal{V}$ and integrating over domain $\mathcal{D}$, hence it is possible to write

$$\int_{\mathcal{D}} w \, \nabla^2 u \, \mathrm{d}v = \int_{\mathcal{D}} w \, s \, \mathrm{d}v \ . \tag{11}$$

Knowing from the product rule that

$$\nabla \cdot (w \nabla u) = \nabla w \cdot \nabla u + w \nabla^2 u \ , \tag{12}$$

$$w \nabla^2 u = \nabla \cdot (w \nabla) - \nabla w \cdot \nabla u \tag{13}$$

it is possible to rewrite (11) as

$$\int_{\mathcal{D}} \left[\nabla \cdot (w \nabla u) - \nabla w \nabla u\right] \mathrm{d}v = \int_{\mathcal{D}} w \, s \, \mathrm{d}v \tag{14}$$

and using divergence theorem

$$-\int_{\mathcal{D}} \nabla w \cdot \nabla u \, \mathrm{d}v + \int_{\mathcal{S}} w (\nabla u \cdot \boldsymbol{n}) \, \mathrm{d}a = \int_{\mathcal{D}} w \, s \, \mathrm{d}v \ . \tag{15}$$

As $w = 0$ on $\mathcal{S}_q$, $\mathcal{S} = \mathcal{S}_u \cup \mathcal{S}_q$ and $\nabla u \cdot \boldsymbol{n} = \frac{\partial u}{\partial x} \cdot \boldsymbol{n} = \bar{q}$, (15) can be written as

$$-\int_{\mathcal{D}} \nabla w \cdot \nabla u \, \mathrm{d}v + \int_{\mathcal{S}_q} w \, \bar{q} \, \mathrm{d}a = \int_{\mathcal{D}} w \, s \, \mathrm{d}v \ . \tag{16}$$

According to Lex-Milgram lemma the convergence of (16) can be shown (see e.g. [Evans] or [Downsea Huerta]). The above is referred as the weak form as the required continuity for unknown function u is reduced by one in comparison to strong form.

With the derived weak form it is possible use the finite element method to solve given problem. First step is to discretize the domain $\mathcal{D}$ into finite element $\mathcal{D}^e$ such that

$$\bar{\mathcal{D}} = \bigcup_{e=1}^{n_{el}} \bar{\mathcal{D}}^e \tag{17}$$

and

$$\mathcal{D}^e \cap \mathcal{D}^f = \emptyset \quad \text{for } e \neq f \ . \tag{18}$$

With the definitions in Eq. 17 the functional spaces can be approximated as

$$\mathcal{V}^h = \{w \in H^1(\mathcal{D}) \mid w|_{\mathcal{S}^e} \in \mathcal{P}_m(\mathcal{D}^e) \, \forall e \text{ and } w = 0 \text{ on } \mathcal{S}_u\} \ , \tag{19}$$

$$\mathcal{W}^h = \{u \in H^1(\mathcal{D}) \mid u|_{\mathcal{S}^e} \in \mathcal{P}_m(\mathcal{D}^e) \, \forall e \text{ and } u = \bar{u} \text{ on } \mathcal{S}_u\} \ . \tag{20}$$

The approximation of the position vectors

$$\boldsymbol{x} \approx \boldsymbol{x}^h = \boldsymbol{N}^e \boldsymbol{x}^e, \quad \boldsymbol{x}^h \in \mathcal{D}^e \subset \mathcal{S}^h \ , \tag{21}$$

H where

$$\boldsymbol{N}^e = [N_1\mathbf{1}, N_2\mathbf{1}, ..., N_{n_n^e}\mathbf{1}] \ , \tag{22}$$

where $\boldsymbol{N}^e$ denotes the $(d \times d\, n_n^e)$ array where n_n^e denotes number of element nodes, d number of degrees of freedom for given node and N_i nodal shape functions. For potential function the discretization can be written as

$$u(\boldsymbol{x}) \approx u^h(\boldsymbol{x}) = \sum_{k=1}^{n_n^e} N_k^e u_k^e = \boldsymbol{N}^e \boldsymbol{u}^e \ . \tag{23}$$

Using Galerkin formulation the arbitrary test functions, w^h can be defined as

$$w^h \in \mathcal{V}^h := \underset{e=1...n_{el}}{\text{span}} \ \{\boldsymbol{N}^e\} \ . \tag{24}$$

With Eqs 22-24 the discretized system of equations can be written as

$$\underbrace{-\sum_{e=1}^{n_{\text{el}}} \int_{\mathcal{D}^e} \nabla N_a \cdot \nabla N_b \, \mathrm{d}v}_{\boldsymbol{K}} \ u_b^e = \underbrace{\sum_{e=1}^{n_{\text{el}}} \int_{\mathcal{D}^e} N_b \, s \, \mathrm{d}v}_{\boldsymbol{f}} - \sum_{e=1}^{n_{\text{el}}^b} \int_{\mathcal{S}_q^e} N_b \, \bar{q} \, \mathrm{d}a \ , \tag{25}$$

where n_{el}^b denotes number of the elements laying on the boundary where the Neumann boundary conditions are prescribed. The system of linear equations required to solve the model problem from Eq. 8 can be written in a compact form as

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f} \ , \tag{26}$$

where the elemental contributions are defined as

$$\boldsymbol{K}_{ab}^e = -\int_{\mathcal{D}^e} \nabla N_a \cdot \nabla N_b \, \mathrm{d}v \ , \tag{27}$$

$$\boldsymbol{f}_b^e = \int_{\mathcal{D}^e} N_b \ s \, \mathrm{d}v \ . \tag{28}$$

To evaluate the integrals in (27)-(28) the isoparametric concept is used. The integration domain is transferred to the parametric coordinate system. For instance considering 1D integral, it is possible to write

$$\int_{x_i}^{x_{i+1}} f(x) \, \mathrm{d}x = \int_{-1}^{1} f(x(\xi)) \, |\boldsymbol{J}| \, \mathrm{d}\xi = \int_{-1}^{1} f(x(\xi)) \frac{\mathrm{d}x}{\mathrm{d}\xi} \, \mathrm{d}\xi \ , \tag{29}$$

where $|\boldsymbol{J}|$ defines the determinant of the Jactobian. In 1D case it is simply the parametric derivative. Note that the change of the domain of integration are achieved by

$$\mathrm{d}x = \frac{\mathrm{d}x}{\mathrm{d}\xi} \mathrm{d}\xi \ . \tag{30}$$

For 2D case the Jacobian can be defined as

$$\boldsymbol{J} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \ . \tag{31}$$

For instance to compute $J_{11} = \frac{\partial x}{\partial \xi}$ for rectangular element with linear shape functions, it is possible to write

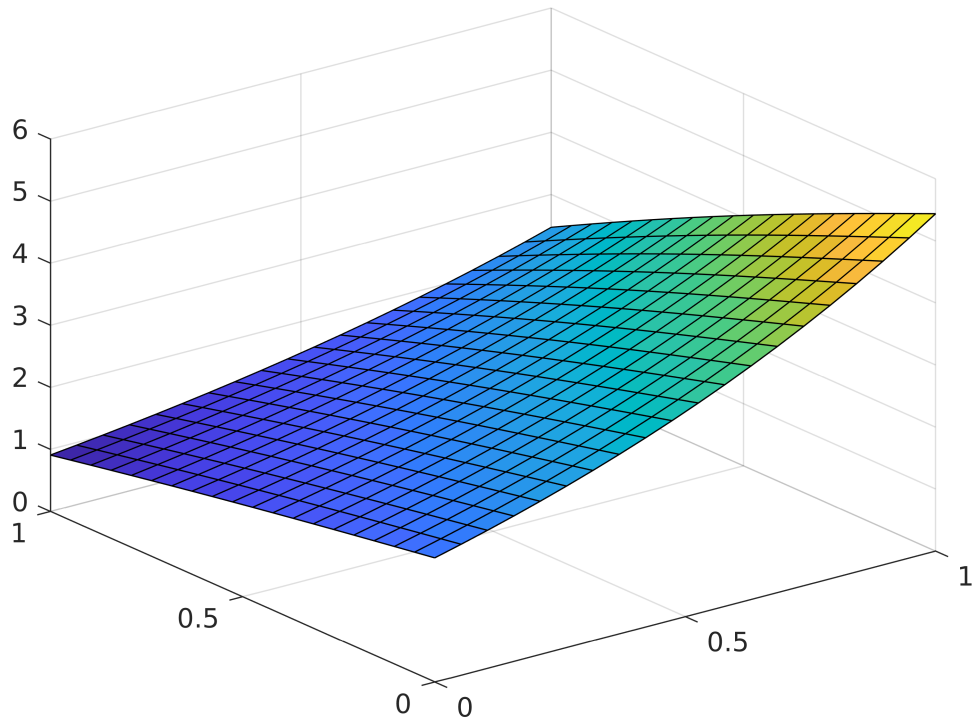
$$\frac{\partial}{\partial \xi} x(\xi, \eta) = \frac{\partial N_1}{\partial \xi} x_1 + \frac{\partial N_2}{\partial \xi} x_2 + \frac{\partial N_3}{\partial \xi} x_3 + \frac{\partial N_4}{\partial \xi} x_4 \ . \tag{32}$$

Additionally, to evaluate (27) it is necessary to calculate the $\frac{\partial N}{\partial x}$ and $\frac{\partial N}{\partial y}$ which can be achieved by evaluating following system of equations

$$\begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \begin{bmatrix} \frac{\partial N}{\partial x} \\ \frac{\partial N}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial N}{\partial \xi} \\ \frac{\partial N}{\partial \eta} \end{bmatrix} \tag{33}$$

and hence

$$\begin{bmatrix} \frac{\partial N}{\partial x} \\ \frac{\partial N}{\partial y} \end{bmatrix} = \boldsymbol{J}^{-1} \begin{bmatrix} \frac{\partial N}{\partial \xi} \\ \frac{\partial N}{\partial \eta} \end{bmatrix} \ . \tag{34}$$



Figuur 1: Exact solution to the problem.

With (29)-(34) and using Gaussian quadrature the contributions to the system matrices $\boldsymbol{K}$ and $\boldsymbol{f}$ from (27)-(28) for 2D case can be written as

$$\boldsymbol{K}^e_{ab} = -\int_{\mathcal{D}^e} \nabla N_a \cdot \nabla N_b \, \mathrm{d}v = -\sum_{\mathrm{qp}=1}^{n_{qp}} \left[\frac{\partial N_a(\xi_{\mathrm{qp}})}{\partial x} \frac{\partial N_b(\xi_{\mathrm{qp}})}{\partial x} + \frac{\partial N_a(\xi_{\mathrm{qp}})}{\partial y} \frac{\partial N_b(\xi_{\mathrm{qp}})}{\partial y} \right] \det \mathbf{J}(\xi_{\mathrm{qp}}) \, w_{\mathrm{qp}} \ , \tag{35}$$

$$\boldsymbol{f}^e_b = \int_{\mathcal{D}^e} N_b \ s \, \mathrm{d}v = \sum_{\mathrm{qp}=1}^{n_{qp}} N_b(\xi_{\mathrm{qp}}) \, s(\xi_{\mathrm{qp}}) \, \det\mathbf{J}\,(\xi_{\mathrm{gp}}) \ w_{\mathrm{qp}} \ . \tag{36}$$

The elemental contributions must be organized accordingly to the connectivity for given discretization. To solve given problem it is necessary to apply the boundary conditions. While Neumann BCs are applied accordingly as show in integral (25) for the respective portion of the body and thereby contributions to given elements, the Dirichlet boundary condition are applied slightly different way. In literature can be seen different approaches (see e.g. Downsea Huerta]). One of the ideas is for the nodes where the Dirichlet boundary conditions are applied respective rows of $\boldsymbol{K}$ becomes zero except the diagonal term where it becomes one and respective elements of $\boldsymbol{f}$ becomes the prescribed values of boundary conditions.

Task

Consider following problem

$$\begin{cases} \nabla^2 u = 2e^{x-y} & \text{in } (x,y) \in \mathcal{D} \ , \\ u = e^{x-y} + e^x \cos y & \text{on } (x,y) \in \mathcal{S} \ , \end{cases} \tag{37}$$

where

$$\mathcal{D} \cup \mathcal{S} = \{(x,y) \mid 0 \le x \le 1, \, 0 \le y \le 1\} \ . \tag{38}$$

The exact solution can be written as

$$u = e^{x-y} + e^x \cos y \ , \tag{39}$$

which can be seen in Fig. 1.

Your tasks:

- Implementation of FEM solver for Poisson problem in (8):
  - Based on provided code for integration of the function (volume under the plot) implement calculation of (35) and (36)
  - Assemble the system using the provided mesh. Here the structured rectangular mesh is used with linear shape functions.
- Validate the solver by means of provided model problem with exact solution in (37).
- Extend the existing code to order of approximation polynomial up $p = 5$
- Evaluature the resulting h- and p- refinemets L^2-error

Nonlinear solver

Consider the Neo-Hookean material model defined as

$$\boldsymbol{\sigma} = \frac{\mu}{J}(\boldsymbol{b} - \boldsymbol{I}) + \frac{\Lambda}{J}\ln J\boldsymbol{I} \tag{40}$$

with

$$\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^T \qquad \boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F} \tag{41}$$

The expression for the compliance matrix can be defined as

$$\mathbb{C} = 4\frac{\partial^2 W}{\partial \boldsymbol{C}\partial \boldsymbol{C}} = 2\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{C}} \tag{42}$$

where W denotes strain density function, $\boldsymbol{C}$ is the left Cauchy-Green tensor and $\boldsymbol{S}$ is the second stress tensor. Based on (40) and (42) derive the explicit expression for compliance matrix.